

The Galactic Canvas

An Interactive Storytelling Journey: Milky Way → Solar System → Sun → Planets

Research Compilation Document — with Answers

Project Overview

The Galactic Canvas is an interactive storytelling website that takes the viewer on a visual and narrative journey — beginning with the Milky Way galaxy, zooming into our Solar System, moving toward the Sun, and finally exploring each planet in detail. The site is designed as a scroll- or zoom-based experience, guiding users from the largest cosmic scale down to individual planetary facts.

Section 1: Purpose, Audience & Rationale

Why Are We Using This Website?

This website is being created to make astronomy engaging and easy to understand through visual storytelling instead of plain text or static images. Space is a subject that is naturally vast and hard to grasp — by using an interactive zoom-based journey (Milky Way → Solar System → Sun → Planets), the site turns abstract scale and distance into something viewers can experience step by step. It combines education with an immersive, cinematic feel, helping users retain information better than a traditional textbook format, while also demonstrating skills in web design, interactivity, and data storytelling.

- To simplify a complex topic (the scale of the universe) through visual, interactive design
- To create an engaging, memorable learning experience rather than passive reading
- To demonstrate how storytelling techniques (zoom, scroll, narrative flow) can be used to teach science
- To showcase creative and technical skills — design, animation, and content research combined

Who Are We Using This Website For? (Target Audience)

- Students (school/college level) who want an engaging way to learn about space and astronomy
- Teachers/educators looking for a visual aid to explain scale, the Solar System, and planets in class
- Space enthusiasts and curious learners who enjoy interactive, self-paced exploration
- Evaluators/examiners of this assignment, who will assess both content accuracy and interactive design
- General website visitors with little prior astronomy knowledge — the content is written to be accessible, not overly technical

Why This Topic Was Chosen (Milky Way / Space)

- Space is a universally fascinating topic with strong visual potential, making it ideal for interactive storytelling
- It naturally supports a 'zoom-in' narrative structure — galaxy to star to planets — giving the project a clear beginning, middle, and end
- Reliable data (distances, sizes, compositions) is well-documented, making research and fact-checking straightforward
- The topic allows use of striking visuals, animations, and comparisons (scale, distance, temperature) that keep users engaged
- It has strong educational value, making the project useful beyond just an assignment — it can genuinely teach people something
- Personal interest in space and astronomy made the research and design process more motivating and enjoyable

Section 2: Research Questions — Answered

Stage 1 — The Milky Way (Establishing Shot)

Q: What is the Milky Way, and what type of galaxy is it (structure-wise)?

A: The Milky Way is the galaxy that contains our Solar System. Structurally, it is classified as a barred spiral galaxy, meaning it has a central bar-shaped core with spiral arms winding outward.

Q: How big is the Milky Way (diameter, number of stars)?

A: The Milky Way is roughly 100,000–120,000 light-years in diameter and contains an estimated 100–400 billion stars.

Q: Where is our galaxy located in the universe, and what galaxies are nearby?

A: The Milky Way is part of a small cluster of galaxies called the Local Group. Its nearest large neighbor is the Andromeda Galaxy, along with smaller satellite galaxies like the Large and Small Magellanic Clouds.

Q: Who discovered/named the Milky Way, and how did ancient cultures interpret it?

A: The Milky Way has been observed since ancient times by cultures worldwide, who described it in myths as a river, road, or trail of light. Galileo Galilei was the first to confirm in 1610, using a telescope, that the band of light was made up of countless individual stars. The name comes from the Latin 'Via Lactea,' translating the Greek description of a 'milky circle.'

Q: How did the Milky Way form, and how old is it?

A: The Milky Way formed roughly 13.6 billion years ago through the gravitational collapse and merging of gas clouds and smaller proto-galactic structures shortly after the Big Bang.

Q: What are the main parts of the galaxy (core, spiral arms, halo, disk)?

A: The Milky Way consists of a central bulge/core, spiral arms extending outward containing most visible stars, a flat rotating disk, and a surrounding halo of older stars, gas, and dark matter.

Q: Why does it look like a band of light across the night sky from Earth?

A: Because our Solar System sits inside the galaxy's flat disk, when we look along the plane of the disk at night, we see the combined light of countless distant stars packed closely together, appearing as a hazy band.

Q: What is at the center of the Milky Way (Sagittarius A*)?

A: At the center lies Sagittarius A*, a supermassive black hole with a mass about 4 million times that of the Sun, around which the entire galaxy rotates.

Stage 2 — The Solar System (Zooming In)

Q: What is the Solar System, and where does it sit within the Milky Way?

A: The Solar System is the Sun and everything gravitationally bound to it — planets, moons, asteroids, and comets. It is located in the Orion Arm, one of the Milky Way's spiral arms, about 26,000 light-years from the galactic center.

Q: How did the Solar System form (nebular hypothesis)?

A: According to the nebular hypothesis, the Solar System formed about 4.6 billion years ago when a giant molecular cloud of gas and dust collapsed under gravity, flattening into a spinning disk. The Sun

formed at the dense center, while the remaining material clumped together to form planets, moons, and other bodies.

Q: How many objects make up the Solar System (planets, dwarf planets, moons, asteroids, comets)?

A: The Solar System includes 8 planets, 5 officially recognized dwarf planets, over 300 known moons, more than a million asteroids larger than 1 km, and countless comets.

Q: Why do planets orbit the Sun the way they do?

A: Planets orbit the Sun because of the balance between the Sun's gravitational pull and the planet's forward momentum — gravity constantly pulls planets inward while their motion carries them forward, resulting in a stable elliptical orbit.

Q: What holds the Solar System together?

A: The Sun's immense gravity, arising from its huge mass (99.86% of the Solar System's total mass), holds all planets, moons, and smaller bodies in orbit around it.

Q: How long does it take the Solar System to orbit the galactic center?

A: The Solar System takes approximately 230 million years to complete one orbit around the Milky Way's center — a period often called a 'galactic year.'

Stage 3 — The Sun (The Zoom Target)

Q: What is the Sun made of, and what type of star is it?

A: The Sun is made of roughly 73% hydrogen and 25% helium (plus trace heavier elements) and is classified as a G-type main-sequence star, commonly nicknamed a 'yellow dwarf.'

Q: How does the Sun produce energy (nuclear fusion)?

A: The Sun produces energy through nuclear fusion in its core, where immense pressure and heat fuse hydrogen atoms into helium, releasing enormous amounts of energy as light and heat.

Q: How old is the Sun, and what is its life cycle/future?

A: The Sun is about 4.6 billion years old and is roughly midway through its main-sequence lifespan, expected to last about 10 billion years in total. It will eventually expand into a red giant before shrinking into a white dwarf.

Q: How hot is the Sun (core vs. surface)?

A: The Sun's core reaches about 15 million°C, while its visible surface (photosphere) is much cooler, at around 5,500°C.

Q: Why is the Sun important for life on Earth?

A: The Sun provides the light and heat that drive Earth's climate, weather, and water cycle, and its energy powers photosynthesis, forming the base of nearly all food chains on Earth.

Q: What are solar phenomena (sunspots, solar flares, solar wind)?

A: Sunspots are cooler, darker regions on the Sun's surface caused by magnetic activity. Solar flares are sudden bursts of intense energy and radiation from the Sun's surface. Solar wind is a continuous stream of charged particles flowing outward from the Sun into space.

Stage 4 — Each Planet (Answered per Planet)

The same nine template questions are answered below for each of the eight planets, covering composition, orbit, rotation, size, moons, atmosphere, uniqueness, discovery, and exploration history.

Mercury

Q: What is the planet made of (rocky / gas / ice giant)?

A: Rocky terrestrial planet with a large metallic core (about 70% metal, 30% silicate rock).

Q: How far is it from the Sun, and how long is its orbit (year length)?

A: 57.9 million km from the Sun; completes one orbit (year) in 88 Earth days.

Q: How long is a day on this planet (rotation period)?

A: One rotation takes 59 Earth days.

Q: What is its size compared to Earth?

A: About 38% of Earth's diameter (4,879 km) — the smallest planet in the Solar System.

Q: Does it have moons — how many, and any notable ones?

A: No moons.

Q: What is its atmosphere like, and surface temperature?

A: Virtually no atmosphere (only a thin exosphere); temperatures swing from about -180°C at night to 430°C during the day.

Q: Why is this planet unique (e.g., Saturn's rings, Mars's red color, Venus's greenhouse effect)?

A: It is the smallest and fastest-orbiting planet, heavily cratered, and experiences the most extreme temperature swings of any planet.

Q: Who discovered it, or when was it first observed/visited by spacecraft?

A: Known since ancient times; first visited by NASA's Mariner 10 spacecraft (1974–75).

Q: Has it been explored by any missions — what did we learn?

A: Explored by Mariner 10 and NASA's MESSENGER orbiter (2011–2015); the BepiColombo mission is currently en route.

Venus

Q: What is the planet made of (rocky / gas / ice giant)?

A: Rocky terrestrial planet with a thick atmosphere of carbon dioxide and clouds of sulfuric acid.

Q: How far is it from the Sun, and how long is its orbit (year length)?

A: 108.2 million km from the Sun; completes one orbit in 225 Earth days.

Q: How long is a day on this planet (rotation period)?

A: One rotation takes 243 Earth days, and it rotates backward (retrograde) compared to most planets.

Q: What is its size compared to Earth?

A: About 95% of Earth's diameter (12,104 km) — very close in size to Earth.

Q: Does it have moons — how many, and any notable ones?

A: No moons.

Q: What is its atmosphere like, and surface temperature?

A: Thick CO₂ atmosphere causing a runaway greenhouse effect; surface temperature averages about 465°C, the hottest of any planet.

Q: Why is this planet unique (e.g., Saturn's rings, Mars's red color, Venus's greenhouse effect)?

A: It is the hottest planet despite not being closest to the Sun, and it uniquely rotates in the opposite direction to its orbit.

Q: Who discovered it, or when was it first observed/visited by spacecraft?

A: Known since antiquity as the 'morning/evening star'; first flown past by Mariner 2 (1962).

Q: Has it been explored by any missions — what did we learn?

A: Explored by the Soviet Venera program (including the first landing in 1970), NASA's Magellan orbiter, and Japan's Akatsuki mission.

Earth

Q: What is the planet made of (rocky / gas / ice giant)?

A: Rocky terrestrial planet with a layered structure: core, mantle, and crust.

Q: How far is it from the Sun, and how long is its orbit (year length)?

A: 149.6 million km from the Sun; completes one orbit in 365.25 days.

Q: How long is a day on this planet (rotation period)?

A: One rotation takes 24 hours.

Q: What is its size compared to Earth?

A: 12,742 km diameter — used as the reference point for comparing other planets.

Q: Does it have moons — how many, and any notable ones?

A: One moon: the Moon.

Q: What is its atmosphere like, and surface temperature?

A: Nitrogen-oxygen atmosphere; average surface temperature about 15°C; liquid water covers roughly 71% of the surface.

Q: Why is this planet unique (e.g., Saturn's rings, Mars's red color, Venus's greenhouse effect)?

A: It is the only known planet to support life, protected by a magnetic field that shields it from harmful solar radiation.

Q: Who discovered it, or when was it first observed/visited by spacecraft?

A: Our home planet — studied continuously throughout human history.

Q: Has it been explored by any missions — what did we learn?

A: Explored extensively via satellites, the International Space Station, and countless Earth-observation missions.

Mars

Q: What is the planet made of (rocky / gas / ice giant)?

A: Rocky terrestrial planet with an iron-oxide-rich surface, giving it a reddish color.

Q: How far is it from the Sun, and how long is its orbit (year length)?

A: 227.9 million km from the Sun; completes one orbit in 687 Earth days.

Q: How long is a day on this planet (rotation period)?

A: One rotation takes about 24.6 hours, very close to an Earth day.

Q: What is its size compared to Earth?

A: About 53% of Earth's diameter (6,779 km).

Q: Does it have moons — how many, and any notable ones?

A: Two small moons: Phobos and Deimos.

Q: What is its atmosphere like, and surface temperature?

A: Thin carbon dioxide atmosphere; average temperature around -63°C.

Q: Why is this planet unique (e.g., Saturn's rings, Mars's red color, Venus's greenhouse effect)?

A: Known as the 'Red Planet,' it shows strong evidence of ancient rivers and lakes, hosts the largest volcano in the Solar System (Olympus Mons), and is a leading target for future human exploration.

Q: Who discovered it, or when was it first observed/visited by spacecraft?

A: Known since antiquity; first successfully flown past by Mariner 4 (1965).

Q: Has it been explored by any missions — what did we learn?

A: Explored by Viking landers, numerous orbiters, and rovers including Curiosity and Perseverance.

Jupiter

Q: What is the planet made of (rocky / gas / ice giant)?

A: Gas giant composed mainly of hydrogen and helium, with no solid surface.

Q: How far is it from the Sun, and how long is its orbit (year length)?

A: 778.5 million km from the Sun; completes one orbit in 11.9 years.

Q: How long is a day on this planet (rotation period)?

A: One rotation takes just 9.9 hours — the fastest rotation of any planet.

Q: What is its size compared to Earth?

A: 139,820 km diameter — the largest planet in the Solar System, about 11 times Earth's diameter.

Q: Does it have moons — how many, and any notable ones?

A: 95 known moons, including the four large Galilean moons: Io, Europa, Ganymede, and Callisto.

Q: What is its atmosphere like, and surface temperature?

A: Hydrogen/helium atmosphere featuring the Great Red Spot, a giant, centuries-old storm, and distinct banded cloud patterns.

Q: Why is this planet unique (e.g., Saturn's rings, Mars's red color, Venus's greenhouse effect)?

A: It is the largest planet, has the strongest planetary magnetic field, and even has a faint ring system.

Q: Who discovered it, or when was it first observed/visited by spacecraft?

A: Known since antiquity; Galileo Galilei discovered its four largest moons in 1610.

Q: Has it been explored by any missions — what did we learn?

A: Explored by Pioneer 10/11, Voyager 1 and 2, the Galileo orbiter, and the ongoing Juno mission.

Saturn

Q: What is the planet made of (rocky / gas / ice giant)?

A: Gas giant composed mainly of hydrogen and helium.

Q: How far is it from the Sun, and how long is its orbit (year length)?

A: 1,434 million km from the Sun; completes one orbit in 29.5 years.

Q: How long is a day on this planet (rotation period)?

A: One rotation takes about 10.7 hours.

Q: What is its size compared to Earth?

A: 116,460 km diameter — the second-largest planet, about 9.4 times Earth's diameter.

Q: Does it have moons — how many, and any notable ones?

A: 146 known moons, notably Titan (with a thick atmosphere and methane lakes) and Enceladus (with icy geysers).

Q: What is its atmosphere like, and surface temperature?

A: Hydrogen/helium atmosphere; famous for its extensive, bright ring system made of ice and rock particles.

Q: Why is this planet unique (e.g., Saturn's rings, Mars's red color, Venus's greenhouse effect)?

A: Its iconic rings make it instantly recognizable, and it is the least dense planet in the Solar System — it would float in water.

Q: Who discovered it, or when was it first observed/visited by spacecraft?

A: Known since antiquity; its rings were correctly identified by Christiaan Huygens in 1655 after Galileo's earlier, unclear observations.

Q: Has it been explored by any missions — what did we learn?

A: Explored by Pioneer 11, Voyager 1 and 2, and extensively by the Cassini-Huygens mission (2004–2017).

Uranus

Q: What is the planet made of (rocky / gas / ice giant)?

A: Ice giant made of water, ammonia, and methane ices along with hydrogen and helium.

Q: How far is it from the Sun, and how long is its orbit (year length)?

A: 2,871 million km from the Sun; completes one orbit in 84 years.

Q: How long is a day on this planet (rotation period)?

A: One rotation takes about 17.2 hours.

Q: What is its size compared to Earth?

A: 50,724 km diameter — about 4 times Earth's diameter.

Q: Does it have moons — how many, and any notable ones?

A: 27 known moons, including Titania and Oberon.

Q: What is its atmosphere like, and surface temperature?

A: Hydrogen/helium/methane atmosphere giving it a pale blue-green color; extremely cold temperatures.

Q: Why is this planet unique (e.g., Saturn's rings, Mars's red color, Venus's greenhouse effect)?

A: It rotates almost on its side, with an axial tilt of about 98°, likely the result of an ancient massive collision.

Q: Who discovered it, or when was it first observed/visited by spacecraft?

A: Discovered by William Herschel in 1781 — the first planet found using a telescope.

Q: Has it been explored by any missions — what did we learn?

A: Visited only once, by Voyager 2 during its 1986 flyby; no dedicated mission has been sent since.

Neptune

Q: What is the planet made of (rocky / gas / ice giant)?

A: Ice giant with a composition similar to Uranus: water, ammonia, and methane ices plus hydrogen and helium.

Q: How far is it from the Sun, and how long is its orbit (year length)?

A: 4,495 million km from the Sun; completes one orbit in 165 years.

Q: How long is a day on this planet (rotation period)?

A: One rotation takes about 16.1 hours.

Q: What is its size compared to Earth?

A: 49,244 km diameter — slightly smaller than Uranus but more massive.

Q: Does it have moons — how many, and any notable ones?

A: 14 known moons, notably Triton, which orbits in the opposite direction to Neptune's rotation.

Q: What is its atmosphere like, and surface temperature?

A: Hydrogen/helium/methane atmosphere; home to the strongest winds in the Solar System, reaching about 2,100 km/h.

Q: Why is this planet unique (e.g., Saturn's rings, Mars's red color, Venus's greenhouse effect)?

A: Known for its deep blue color and for being the only planet discovered through mathematical prediction rather than direct observation.

Q: Who discovered it, or when was it first observed/visited by spacecraft?

A: Discovered in 1846 by Johann Galle, based on calculations by Urbain Le Verrier predicting its position from Uranus's orbital irregularities.

Q: Has it been explored by any missions — what did we learn?

A: Visited only once, by Voyager 2 during its 1989 flyby; no dedicated mission has been sent since.

Bonus — Interactive / Design Research

Q: How have other space websites/apps handled zoom-scroll storytelling (for UX inspiration)?

A: Many space-education sites use scrollytelling techniques — where scrolling triggers animations, zooms, or transitions — paired with parallax backgrounds to create a sense of depth and movement through space, keeping the scale of the universe visually intuitive.

Q: What tools/libraries support parallax or scroll-based zoom effects (Three.js, GSAP ScrollTrigger)?

A: Three.js can render 3D scenes (like a rotating galaxy or planets) directly in the browser, while GSAP's ScrollTrigger plugin ties animations and camera movement to scroll position, making a smooth zoom-in narrative achievable with web technologies.

Q: What accurate scale or distance comparisons could make the zoom feel real (since true-to-scale space is mostly empty)?

A: Since real astronomical distances are too vast to represent literally on screen, using relatable comparisons (e.g., 'if the Sun were the size of a basketball, Earth would be a peppercorn 30 meters away') or compressed, non-linear scale transitions helps convey scale without making empty space feel boring.

Section 3: Quantitative & Qualitative Data

Stage 1 — The Milky Way

Quantitative Data:

- Diameter: ~100,000–120,000 light-years
- Number of stars: 100–400 billion
- Age: ~13.6 billion years
- Solar System's distance from galactic center: ~26,000 light-years
- Thickness of the galactic disk: ~1,000 light-years
- Orbital speed around galactic center: ~828,000 km/h (1 orbit = ~230 million years — a 'galactic year')
- Number of known galaxies in the observable universe: ~2 trillion (estimated)

Qualitative Data:

- Classified as a barred spiral galaxy
- Made of a central bulge, spiral arms, a surrounding halo, and dark matter
- Contains gas, dust, and stellar nurseries where new stars form
- Named 'Milky Way' from its milky band appearance in the night sky (Latin: Via Lactea)
- Part of the 'Local Group' of galaxies, moving toward the Andromeda Galaxy (future collision predicted)

Stage 2 — The Solar System

Quantitative Data:

- Age: ~4.6 billion years
- Number of planets: 8 (plus 5 recognized dwarf planets)
- Diameter of the Solar System (to Oort Cloud): ~2 light-years
- Number of known moons: 300+ across all planets
- Asteroid belt: over 1 million asteroids larger than 1 km

Qualitative Data:

- Formed from a collapsing giant molecular cloud (nebular hypothesis)
- Structure: inner rocky planets, asteroid belt, outer gas/ice giants, Kuiper Belt, Oort Cloud
- Held together by the Sun's gravity
- Planets orbit in roughly the same plane (ecliptic plane) and direction

Stage 3 — The Sun

Quantitative Data:

- Diameter: ~1.39 million km (109x Earth's diameter)
- Mass: 99.86% of the Solar System's total mass
- Surface temperature: ~5,500°C
- Core temperature: ~15 million°C
- Age: ~4.6 billion years; expected lifespan ~10 billion years
- Composition: ~73% hydrogen, ~25% helium

Qualitative Data:

- Classified as a G-type main-sequence star ('yellow dwarf')
- Powered by nuclear fusion (hydrogen fusing into helium)

- Exhibits phenomena like sunspots, solar flares, and solar wind
- Will eventually become a red giant, then a white dwarf

Stage 4 — Planets: Quantitative Reference Table

Planet	Distance from Sun (M km)	Diameter (km)	Day length	Year length	Moons
Mercury	57.9	4,879	59 days	88 days	0
Venus	108.2	12,104	243 days	225 days	0
Earth	149.6	12,742	24 hrs	365.25 days	1
Mars	227.9	6,779	24.6 hrs	687 days	2
Jupiter	778.5	139,820	9.9 hrs	11.9 years	95
Saturn	1,434	116,460	10.7 hrs	29.5 years	146
Uranus	2,871	50,724	17.2 hrs	84 years	27
Neptune	4,495	49,244	16.1 hrs	165 years	14

Qualitative Highlights (for storytelling flavor):

- Mercury — smallest planet, extreme temperature swings, cratered like the Moon
- Venus — hottest planet due to runaway greenhouse effect, rotates backward
- Earth — only known planet with life, liquid water, protective atmosphere
- Mars — 'Red Planet' due to iron oxide dust, evidence of ancient water, target for human exploration
- Jupiter — largest planet, Great Red Spot storm, strong magnetic field
- Saturn — iconic ring system made of ice and rock, least dense planet (would float in water)
- Uranus — rotates on its side (~98° tilt), pale blue-green from methane
- Neptune — strongest winds in the Solar System, deep blue color, discovered through mathematical prediction

Note: Moon counts (especially for Jupiter and Saturn) update occasionally as new moons are confirmed — worth a quick check closer to submission for precision.

Section 4: Project Name

Chosen Name:

The Galactic Canvas

This name reflects the site's core concept — presenting the vastness of the galaxy, the Solar System, and the planets as an unfolding visual and narrative canvas for the viewer to explore.